

Corporate Wallet Digital Twin

Reconstruct the unseen wallet. Decide what matters next.

TEAM: Corporate Wallet Digital Twin

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AS OF 10 AUGUST 2026 | V3.0 | SYN BANK SIMULATION + PUBLIC E1



1,500

SHADOW-WALLET EDGES

100

CLIENT-PRODUCT NETWORKS

12

CAPACITY-AWARE RM ACTIONS

8

POSITIVE-NET-VOI REQUESTS

The decision: allocate twelve RM actions under downside risk

Expected scenario value and lower-tail CVaR are optimized with client, product and sector capacity constraints.

CLIENT / PRODUCT	EXPECTED	DOWNSIDE CVAR	NEED
1. Glencore - Trade finance	R11.1m	R5.0m	100%
2. BHP Group - Trade finance	R4.6m	R2.1m	100%
3. MTN Group - Cross-border FX	R3.7m	R1.7m	100%

A = q x T

Observed = bank share x total wallet

GLENCORE TRADE FINANCE

Entropy-constrained Shadow Wallet

Observed R1.1bnLatent R13.9bnShare 7.2%

256 draws | 5 corridors x 3 anonymous providers | exact median mass balance | SCENARIO, not measured

V3 mechanics reconcile exactly without inventing empirical truth

R0

maximum Shadow Wallet mass-balance error

100

Bayesian change-point series replayed

8 / 8

selected evidence requests have positive net VOI

Claim audit: 0 measured competitor-share claims | 0 causal-value claims | reconstructed leakage remains a modelled verification signal.

RATE / PRIOR SENSITIVITY

Trade Finance remains first-ranked

9/9 benchmark cases | 100% of 10,000 draws

But it is never a majority of the top 10. Cross-border FX is majority-dominant in 79.7% of draws.

DECISION-DIRECTED RAG + GENAI

Score -> value -> approve -> compile -> brief

8 requests | R43.3m net VOI | 0 autonomous retrievals

Evidence is requested only when expected decision value exceeds acquisition cost and latency. The LLM sees a closed claim pack; deterministic fallback remains.

DELIVERED: executed V3 notebook | entitled decision lab | Shadow Wallet | BOCPD signals | robust portfolio | VOI queue | judging deck

External gates: E3 multibank panel, bank-approved economics/infrastructure, live-provider evaluation and supervised RM trial.

Data label: SYN BANK SIMULATION + PUBLIC E1 + REPRESENTATIVE PRIORS - RECONSTRUCTED COMPETITOR FLOWS ARE NOT MEASURED

Code: <https://github.com/ChristopherJohannesKoen/corporate-wallet-digital-twin-hackathon-2026> (private repository - reviewer access required)